

Quantum Mechanics-2

Aug 26

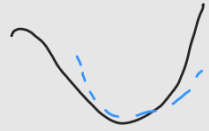
Syllabus overview

- 1) Time independent Perturbation theory
- 2) Atom Structure (H)
- 3) Time dependent perturbation theory.
- 4) Scattering theory

8) Would TPT be applicable to a potential made of two or more known potentials? Something like $U(x) = \frac{1}{2} m \omega^2 x^2 + \text{perturb}$ in a box?

Time independent Perturbation theory

Motivation: consider H.O.



suppose we add a correction:

$$V(x) = \frac{1}{2} m \omega^2 x^2 + g x^3$$

— How does that affect the energies and wavefunctions?

$$E_n^{(0)} = \hbar \omega (n + 1/2) \rightarrow \text{unperturbed energy}$$

General Problem

- given hamiltonian H_0 with known eigenvalues & eigenstates.

$$H_0 |n^{(0)}\rangle = E_n^{(0)} |n^{(0)}\rangle$$

- find approximate eigenstates / values of

$$H = H_0 + V \rightarrow \text{perturbation (sometimes labeled } H')$$

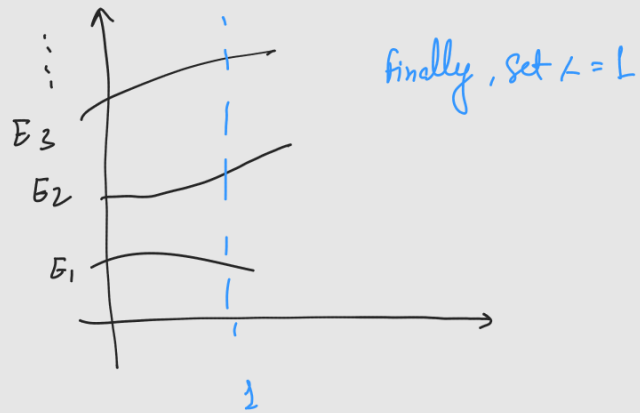
$$H|n\rangle = E_n |n\rangle$$

Method

- introduce $\lambda \in [0, 1]$

$$H(\lambda) = H_0 + \lambda V$$

$$H(\lambda) |n; \lambda\rangle = E_n(\lambda) |n; \lambda\rangle$$



Power Series

$$- E_n(\lambda) = E_n^{(0)} + \lambda \Delta_n^{(1)} + \lambda^2 \Delta_n^{(2)} + \dots$$

$$- |n; \lambda\rangle = |n^{(0)}\rangle + \lambda |n^{(1)}\rangle + \dots$$

$$\text{note: } \lambda \rightarrow 0 : E_n \rightarrow E_n^{(0)}$$

$$|n\rangle = |n^{(0)}\rangle$$

* perturbation depends
on the state of
system before it
was perturbed

Results

- assume non-degenerate ; $E_n^{(0)} \neq E_k^{(0)}$ for $k \neq n$
- normalized $|n^{(0)}\rangle$: $\langle n^{(0)} | n^{(0)} \rangle = 1$

then $\Delta_n^{(1)} = \langle n^{(0)} | V | n^{(0)} \rangle$

$$\Delta_n^{(2)} = \sum_{m \neq n} \frac{|V_{mn}|^2}{E_n^{(0)} - E_m^{(0)}}, \quad V_{mn} = \langle m^{(0)} | V | n^{(0)} \rangle$$

$$|n^{(1)}\rangle = \sum_{m \neq n} |m^{(0)}\rangle \frac{V_{mn}}{E_n^{(0)} - E_m^{(0)}}$$

Quick Example

$$H_0 = \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix}, \quad a < b, \quad |1^{(0)}\rangle = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad |2^{(0)}\rangle = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

$\Rightarrow E_1^{(0)} = a, \quad E_2^{(0)} = b$

perturbation: $V = \begin{pmatrix} c & d \\ d & f \end{pmatrix} \quad c, d, f \in \mathbb{R}$

a) 1st order energy shift of ground state

$$\Delta_1^{(1)} = (1 \ 0) \begin{pmatrix} c & d \\ d & f \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \boxed{c}$$

b) 2nd order energy shift of ground state

$$\Delta_1^{(2)} = \sum_{\substack{m \neq 1 \\ m=2}} \frac{|V_{m1}|^2}{E_1^{(0)} - E_m^{(0)}} = \frac{|V_{21}|^2}{E_1^{(0)} - E_2^{(0)}} = \boxed{\frac{d^2}{a-b}} < 0$$

$\therefore$

$$E_1 \approx \underset{\uparrow}{a} + \underset{\uparrow}{c} + \underset{\uparrow}{\frac{d^2}{a-b}}$$

$\lambda^0 \quad \lambda^1 \quad \lambda^2$

c) 1st order correction to $|1\rangle$

$$|1^{(1)}\rangle = \sum_{m \neq 1} |m^{(0)}\rangle \frac{V_{m1}}{E_1^{(0)} - E_m^{(0)}}$$

$$= |2^{(0)}\rangle \frac{V_{21}}{E_1^{(0)} - E_2^{(0)}}$$

$$= \boxed{\begin{pmatrix} 0 \\ 1 \end{pmatrix} \frac{d}{a-b}}$$

$$\Rightarrow |1\rangle \approx \begin{pmatrix} 1 \\ 0 \end{pmatrix} + \frac{d}{a-b} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$